Coding & Solution

Date	30 June 2026
Team ID	xxxxxx
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/Dimple1603/OptiCrop
Programming Language(s)	Python,JavaScript,HTML5,CSS3
Framework(s) Used	Flask,React.js,Vite
Key Features Implemented	Crop Recommendation, Random Forest Model, Flask API, React UI, Input Validation
Pending / Incomplete Features	Weather API,Cloud Deployment,Mobile App
Setup / Run Instructions	Clone → Install Dependencies → Run Flask →Run React → Predict Crop

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

OptiCrop uses a modular architecture with a React frontend, Flask backend, and a Random Forest machine learning model to provide accurate crop recommendations based on soil and environmental parameters.